3D/N-tier Stacked Memristive Crossbars for Model Confidentiality in Analog In-Memory Computing

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Introduction

- Memristive crossbar array (MCA): analog in-memory computing by driving row voltages and summing column currents; cell conductance encodes weights.
- Parallel dot-products, less data movement; good for ML, neuromorphic.
- 2D to 3D/N-tier: more design freedom, changes compute and exposure.
- 3D/N-tier – security-aware partitioning without changing the compute principle.

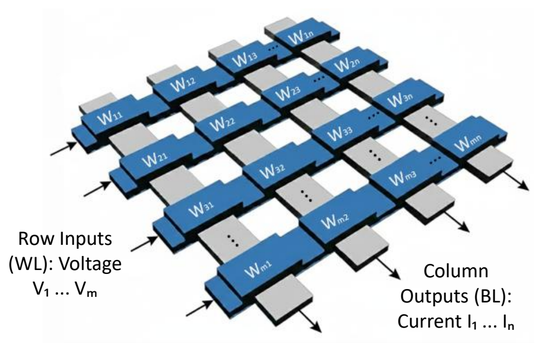

Fig. 1. 2D MCA for analog VMM with cross-point weights W_{ij} implemented in HSPICE as memristor resistance states R_{ij}.

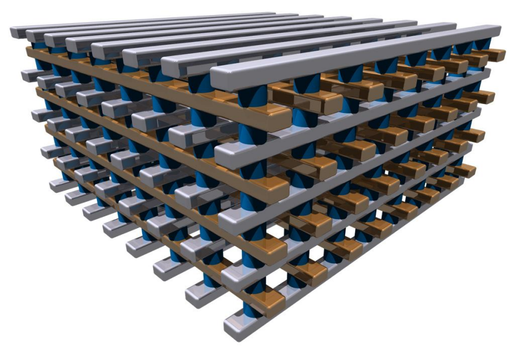

Fig. 2. Illustrative 3D MCA stacking architecture (adapted from a 3D RRAM stack image, Crossbar Inc.).

Problem Statement

- Weights & I/O interface co-located; invasive access can leak model info.
- Need architecture to cut observability, keep compute correct, overhead manageable.
- Confidentiality loss enables IP leakage and model cloning.

Threat Model: Single-Tier Physical Access

- Attacker: probes one exposed tier, applies chosen inputs, and measures WL/BL waveforms and column currents (many queries).
- No access to other tiers; no direct view of full cross-tier aggregation or tier-to-logical mapping (if hidden).
- Baseline Observability for 2D vs 3D:
  - 2D MCA: single layer; WL/BL probing can reveal model parameters.
  - 3D/N-tier MCA: one-tier probe shows partial weights/outputs; full recovery needs multi-tier aggregation.

Proposed Approach: 3D/N-Tier Weight Confidentiality

- Weight matrix split across N tiers; each tier stores only part of parameters.
- Final analog output via cross-tier aggregation; one tier alone hides full dot-product behavior.
- Hidden tier-to-logical mapping; physical nodes do not directly show logical weights.

Designer vs Attacker View

- Designer: Combines tier outputs to recover the correct full result.
- Attacker: Probing one tier shows only a slice, not the full model behavior.
- Result: Physical access to one tier doesn't directly reveal the logical weights

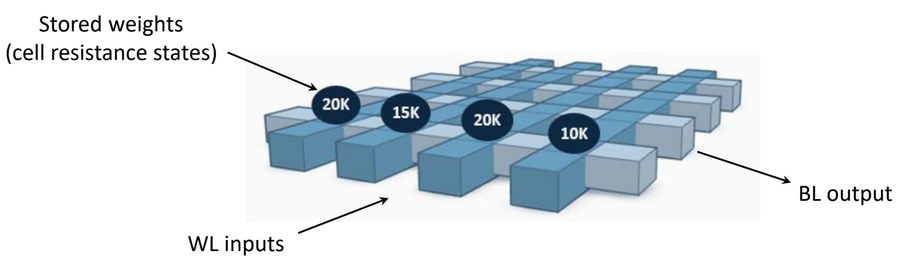

Fig. 3. 2D MCA baseline with unsplit weight layout (illustrative values). A single-layer view exposes the full weight pattern to WL/BL probing.

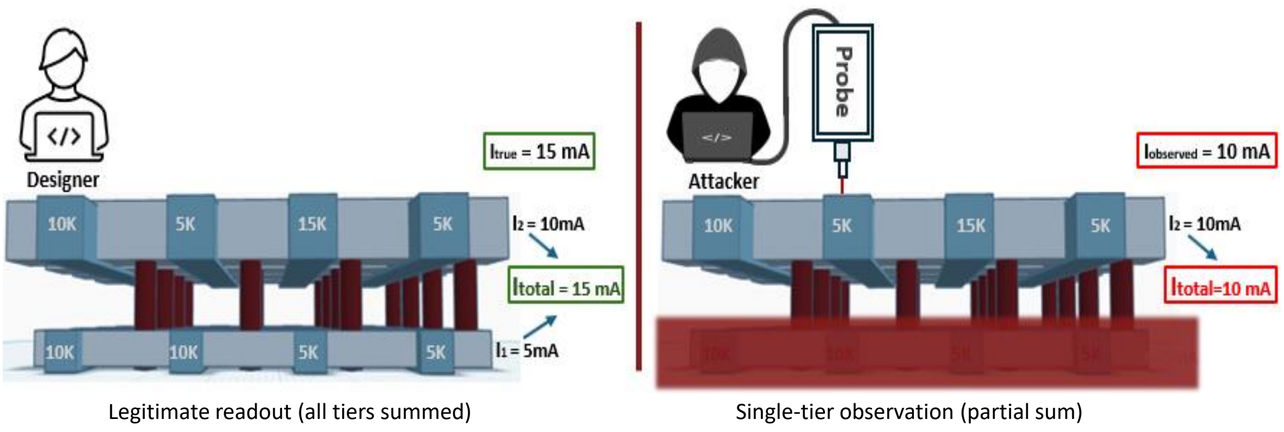

Fig. 4. 3D MCA weight splitting: full-sum readout vs single-slice observation (conceptual values).

Experimental Setup

- HSPICE: TSV-stacked 2-tier MCA (256×128/tier), 1T1R; 2D baselines (256×256 & 256×128) for comparison.
- 45 nm CMOS; WL/BL as per-cell RC ladders (M3–M4 baseline); vertical via as π-model (TSV + bump/pad/RDL lumped).
- Input/sense: 0.1 V row drive with ~150 Ω source; column readout via virtual-ground clamp + C_{LOAD} (≈1 fF) through R_{SUM} = 0.1 Ω.
- Metrics: peak output current, delay (0.9× settling), and average total power.

Findings

- Confidentiality intuition: correct output requires cross-tier summation; a single-tier probe observes only a partial slice.
- Function-matched (256×128): Current gain 1.56 times, Delay 1.51 times, Power 1.57 times (3D vs 2D).
- Cell-count matched (256×256): Current gain 2.70 times, Delay 0.9 times (lower), Power 1.23 times (3D vs 2D).
- Overhead drivers: vertical via parasitics and cross-tier summation network.

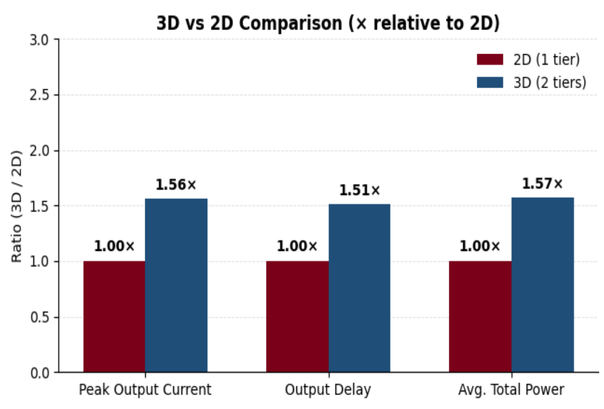

Fig. 5. Function-matched: 256×128 2D (10 kΩ) vs 2-tier 3D 256×128/tier (20 kΩ/tier), 3D/2D ratios.

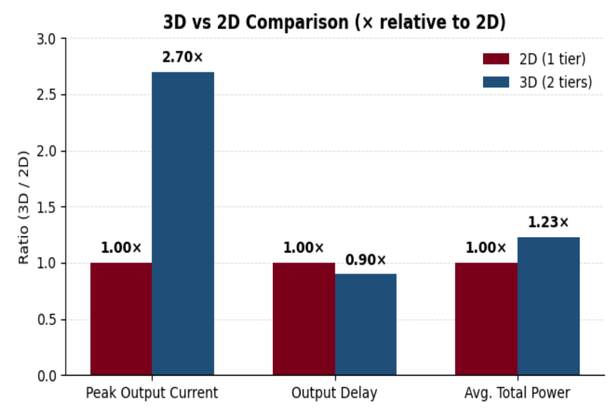

Fig. 6. Cell-count matched: 256×256 2D (10 kΩ) vs 2-tier 3D 256×128/tier (20 kΩ/tier), 3D/2D ratios.

Limitations and Trade-Offs

- More tiers: stronger thermal coupling, harder integration; more interconnect/device variation.
- Vertical interconnects and cross-tier sensing add latency, area, energy.
- More tiers can improve confidentiality under single-tier access, if aggregation and mapping stay hidden.

Conclusion & Future Work

- 3D (2-tier) MCAs improve model confidentiality by limiting single-tier observability.
- Next: leakage vs tiers/connectivity (observable fraction, reconstruction error, query cost), including 3/4/N-tier extensions.
- Co-design: include peripherals, thermal coupling, and variation; add calibration and mapping protection.
- Validation: build small TSV-stacked 3D MCA prototype; confidentiality measurement vs PPA.

References

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